Settle the argument: action and reaction

14 I

DIRECTIONS · Shade the bubble of the person who is right. Use the swap-the-names rule.

ROUND 1

ANA
"The pair forces cancel, so the carts can't move."

BEN
"They act on different objects, so they can't cancel."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"In a truck vs. car crash, the truck pushes harder."

LU
"The forces are equal. The car accelerates more."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"Weight and the table's push are a 3rd-law pair."

RAJ
"Both act on the book. Weight's partner is the book pulling Earth."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

R8 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: force pairs

Language goal: I can **explain** why pair forces never cancel.



A ⇄ B

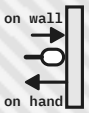
interaction

interacción · interação

two objects push or pull each other

my language: _____

use it: _____



→ ←

force pair

par de fuerzas · par de forças

equal size, opposite direction

my language: _____

use it: _____



force ON the box

F on B

on

sobre · sobre

the object that feels the force

my language: _____

use it: _____



F by A

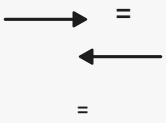
by

por · por

the object that exerts the force

my language: _____

use it: _____



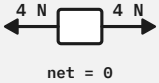
equal

igual · igual

same size

my language: _____

use it: _____



net = 0

+5 − 5 = 0

cancel

cancelar · cancelar

add to zero on ONE object

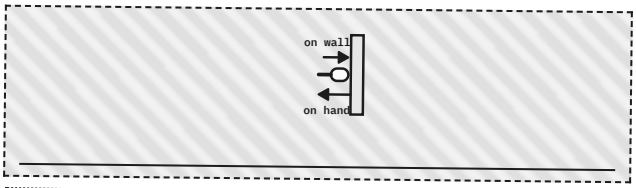
my language: _____

use it: _____

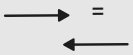
MATCH · WRITE THE WORD FOR EACH PICTURE



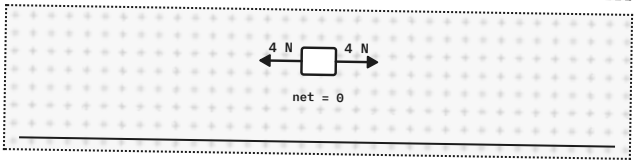
_____



_____



_____



_____

I can point and say all 6 words to a partner □

R8 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Skateboard push-off, in six panels

14 E

DIRECTIONS · Two skaters push off each other. Draw both, with force arrows on EACH skater. Fill the captions.

1 · READY	2 · THE PUSH	3 · ANA
Both at rest. Masses: Ana 40 kg, Ben 80 kg.	Ana pushes Ben with 120 N. Ben pushes Ana with ____ N.	Ana's a = 120 ÷ 40 = ____ m/s².
4 · BEN	**5 · ROLLING APART**	**6 · THE TRUTH**
Ben's a = 120 ÷ 80 = ____ m/s².	Who moves faster? ____ Why?	Equal forces, ____ accelerations.

DRAW · ONE DIAGRAM PER SKATER (only forces ON that skater)

WHY DON'T THE PUSH FORCES CANCEL?

R8 1 2 3

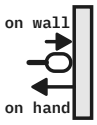
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Talk it through: force pairs

Language goal: I can **swap the names** to find a partner force.

DIRECTIONS · Read your lines aloud. Listen, then write. Switch roles each round.

ROUND 1 · A HAND PUSHES A WALL

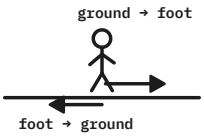


A ASKS "What is the partner force?"

B SAYS "I swap the names: the ____ pushes the ____, in the ____ direction."

A CHECKS "So the forces are ____ in size and act on ____ objects."

ROUND 2 · SWITCH · WALKING

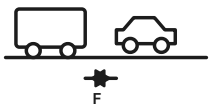


B ASKS "Your foot pushes the ground backward. What pushes you forward?"

A SAYS "The ____ pushes my ____ forward."

B ADDS "I want to add that this is why ____."

ROUND 3 · BOTH · TRUCK MEETS CAR



A "The forces are ____ because ____."

B "But the car's acceleration is ____ because its mass is ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

Third-law forces never cancel because…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R8 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Three third-law myths are on the loose

14 Y

Fill each poster: the truth and evidence that proves it.

WANTED

FOR CRIMES AGAINST PHYSICS

"The bigger object pushes harder."

LAST SEEN: a truck hitting a small car

THE TRUTH:

EVIDENCE:

REWARD: EQUAL FORCES

WANTED

FOR CRIMES AGAINST PHYSICS

"Action and reaction cancel out."

LAST SEEN: two carts pushing apart

THE TRUTH:

EVIDENCE:

REWARD: DIFFERENT OBJECTS

WANTED

FOR CRIMES AGAINST PHYSICS

"Weight and normal force are a pair."

LAST SEEN: a book on a table

THE TRUTH:

EVIDENCE:

REWARD: SWAP THE NAMES

CAPTURED! · WRITE YOUR OWN WANTED NOTICE

A wrong idea about today's physics I once believed:

R8 1 2 3

Pair or not a pair?

14 N

DIRECTIONS · Sort each pair of forces. Third-law pairs act on two different objects.

A hand pushes wall / wall pushes hand	**B** Earth pulls book / table pushes book	**C** foot pushes ground / ground pushes foot	**D** rope pulls box / friction on box
E Earth pulls Moon / Moon pulls Earth	**F** bat pushes ball / ball pushes bat	**G** rocket pushes gas / gas pushes rocket	**H** weight of a lamp / tension in its rope
I cart A pushes B / cart B pushes A	**J** you pull a sled / friction on the sled	**K** magnet pulls clip / clip pulls magnet	**L** swimmer pushes water / water pushes swimmer

3rd-LAW PAIR	NOT A PAIR	NOT SURE YET
A on B ⇄ B on A	same object, or different types	explain below

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

R8 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Comic: every push pushes back

14 CS

YOUR READER
A younger student who thinks only the person who pushes feels a force.
`Stick figures are fine. The physics gets checked.`

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **ORANGE** — Two equal-length force arrows, each labeled "on ____ by ____"
- ☐ ◌ **PURPLE** — Velocity arrows in panel 3, pointing opposite ways
- ☐ ◯ **PENCIL** — "equal and opposite" in a speech bubble

1 · READY *Two skaters stand face to face on ice.*

CAPTION _____

2 · THE PUSH *One skater pushes the other.*

CAPTION _____

3 · BOTH MOVE *Show what happens to both skaters.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **force pair · equal · opposite · on / by**

R8 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · interaction pair

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Interaction pair · par de interacción (ES) · par de interação (PT)



TODAY'S CASE

A skater pushes a wall. The wall pushes the skater.

01 · SKETCH AND LABEL

Sketch the two objects and put one orange force arrow on each receiver. Name the source of each force.

02 · CHANGE THE PICTURE

Redraw the skater alone. Include only the horizontal force acting on the skater. Explain why the pair does not cancel here.

Annotate your sketches. Circle the mark that best shows interaction pair. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

The partner force acts on …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · interaction pair

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Interaction pair · par de interacción (ES) · par de interação (PT)

A · The wall pushes the hand and the hand pushes the wall.

B · Weight and the normal force on a resting book form a third-law pair.

01 · DEFINITION

What does interaction pair mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else